

Course Code: CS3001	Course Name: Computer Networks
Instructor Names: Mr. Muhammad Ali	
Student Roll No:	Section No:

Instructions:

- Return the question paper. There are 4 Questions and written on 5 pages.
- All questions must be answered in answer script and according to the sequence given in the question paper.
- In case of any ambiguity, you may make an assumption. But your assumption should not contradict any statement in the question paper.

Allowed Time: 03 Hours

Max Points: 100 Points

Question # 1:

CLO-02

[04*05= 20 Points]

Part (a) Consider the scenario shown in Figure 1, with four different servers connected to four different clients over four three-hop paths. The four pairs share a common middle hop with a transmission capacity of $R = 400$ Mbps. The four links from the servers to the shared link have a transmission capacity of $R_S = 90$ Mbps. Each of the four links from the shared middle link to a client has a transmission capacity of $R_C = 80$ Mbps. Calculate is the maximum achievable end-end throughput (in Mbps) for each Assuming that the servers are sending at the maximum rate possible, what are the link utilizations for the server links (R_S)? Answer as a decimal of four client-to-server pairs, assuming that the middle link is fairly shared (divides its transmission rate equally)? Assuming that the servers are sending at the maximum rate possible, what are the link utilizations for the client links (R_C)? Answer as a decimal. Assuming that the servers are sending at the maximum rate possible, what is the link utilizations for the shared link (R)? Answer as a decimal

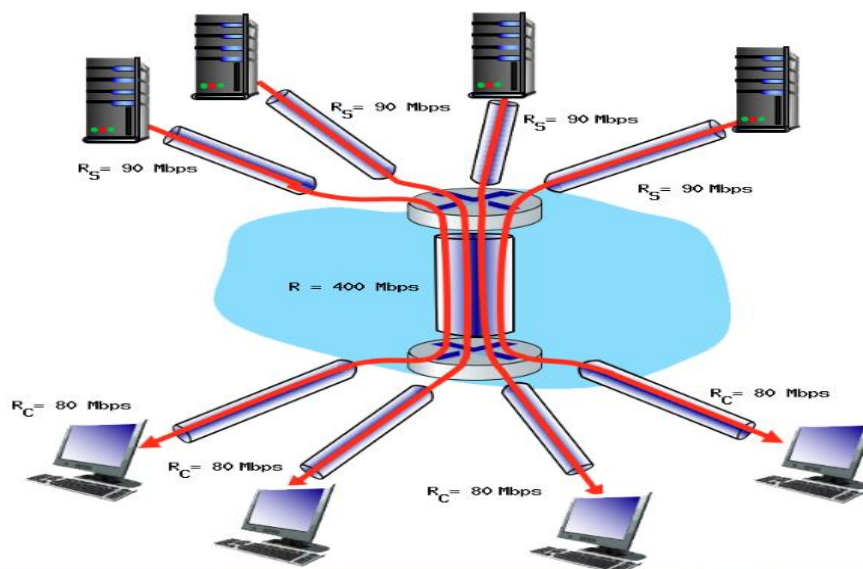


Figure 1.

Part (b) Assume that a user is trying to visit FAST website, but his browser doesn't know the IP address of the website. In this example, examine the difference between an iterative and recursive DNS query with the help of diagram.

Part (c) Consider an HTTP server and client as shown in the figure 2 below. Suppose that the RTT delay between the client and server is 10 msec; the time a server needs to transmit an object into its outgoing link is 0.25 msec; and any other HTTP message not containing an object has a negligible (zero) transmission time. Suppose the client again makes 70 requests, one after the other, waiting for a reply to a request before sending the next request.

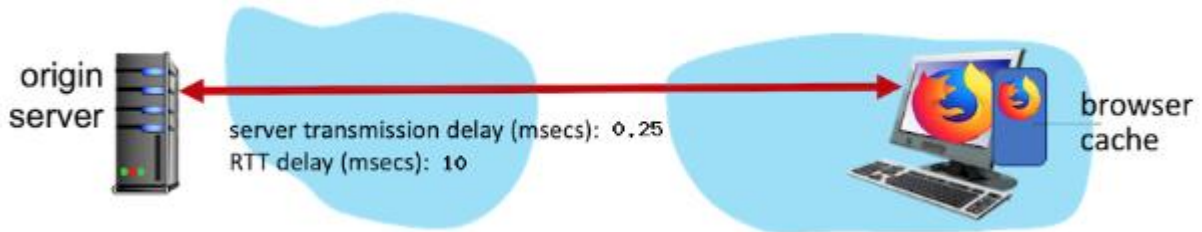


Figure 2

Part (d) Differentiate between client server and P2P network. (Two line answer)

Question # 2:

CLO-03

[2 * 10 = 20 Points]Part

(a) Suppose that TCP's current estimated values for the round trip time (*estimatedRTT*) and deviation in the RTT (*DevRTT*) are 400 msec and 39 msec. Suppose that the next three measured values of the RTT are 240 msec, 350 msec, and 260 msec respectively. Compute TCP's new value of *DevRTT*, *estimatedRTT*, and the TCP timeout value after each of these three measured RTT values is obtained. Use the values of $\alpha = 0.125$, and $\beta = 0.25$. Round your answers to two decimal places after leading zeros. Compute the estimated RTT after the first RTT and the RTT Deviation for the the first RTT? What is the TCP timeout for the first RTT?

Part (b)

Considering the following scenario shown in Figure 02 and briefly answer the questions below.

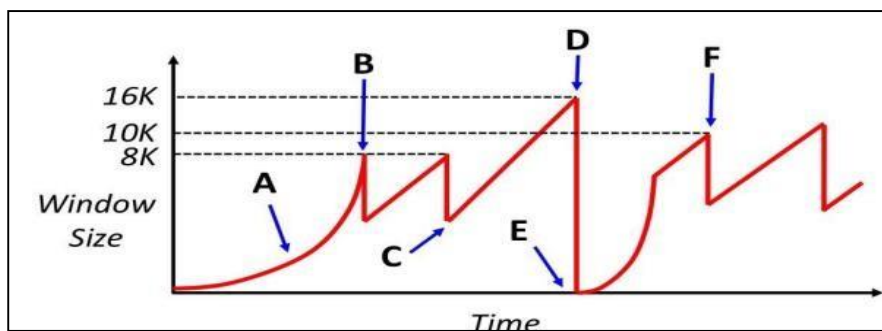


Figure 3

- Explain the event at B which occurs and causes the sender to decrease its window.
- Does the event at B necessitate that the network discarded a packet? Explain.
- Explain the event at D which occurs and causes the sender to decrease its window.
- Does the event at D necessitate that the network discarded a packet? Explain.
- What phenomenon event A represents? Explain.

Question # 3:**CLO-03****[05 * 04 = 20**

Points]Part (a) Host A and B are communicating over a TCP connection; Host B has already received from Host A all bytes up through byte 126. Suppose Host A then sends two segments to Host B back-to-back. The first and second segments contain 80 and 40 bytes of data, respectively. In the first segment the sequence number is 127, the source port number is 302, and destination port number is 80. Host B send an acknowledgment whenever it receives a segment from Host A.

- i. With the help of diagram show how Host A and B establish connection using three-way hand shake.
- ii. In the second segment sent from Host A to B, what are the sequence number, source port number, & destination port number?
- iii. If the second segment arrives before the first segment, in the acknowledgment of the first arriving segment,
- iv. what is the acknowledgement number?
- v. What will be the acknowledgement numbers send by Host B after receiving first segment and second segment in order?

Part (b) What four pieces of information does a DHCP client (Ali's Laptop) obtain from the DHCP server after a successful communication between the Client and Server?

Part (c) Suppose a sender with window size equal to 4 send packets 0,1,2,3 at t=0 and packet 2 is lost, not received at the receiver side. With the help of diagram, show how Go-Back-N algorithm handle this situation

Part (d) Network Congestion is one of the top most research topics in Computer Networks. Explain how TCP congestion control mechanism work and what are its phases?

Part (e) UDP is considered as a faster than TCP. Even though it is faster, there are applications that uses TCP as a transport layer protocol. Briefly explain why TCP is considered over UDP for some applications. Clearly identify few applications that uses TCP

Question # 4:**CLO-03****[04*10 = 40 Points]**

Part (a) NUCES Karachi campus, School of Computing has been assigned the network 208.16.7.0/24 and wants to divide it among the five Departments. Computer Science (CS) needs to accommodate up to 122 hosts, Software Engineering (SE) needs to accommodate up to 58 hosts, Artificial Intelligence & Data Science (AI & DS) needs to accommodate up to 30 hosts, Cyber Security (CY) needs to accommodate up to 12 hosts and Robotics (Robo) needs to accommodate up to 5 hosts. Fill the following table in your answer script with the details of the sub-networks that the NUCES can create to fit its department's needs

Subnet No.	Network Address	Custom Subnet Mask	Usable IP range	Broadcast Address	No. of Hosts
CS					
SE					
AI & DS					
CY					
Robo					

Part (b) Consider the 6-node network shown below, with the given link costs:

A University uses the address block of 192.168.9.0/24 for its network. Now your task is to assign IP addresses to the different department's LAN as shown in network Figure 06. The network has the following addressing requirements.

- When the algorithm converges, what are the distance vectors from router 'Y' to all routers? Write your answer as u,v,w,x,y
- What are the initial distance vectors for router 'X'? Write your answer as u,v,w,x,y and if a distance is ∞ , write 'x'
- The phrase 'Good news travels fast' is very applicable to distance vector routing when link costs decrease; what is the name of the problem that can occur when link costs increase?

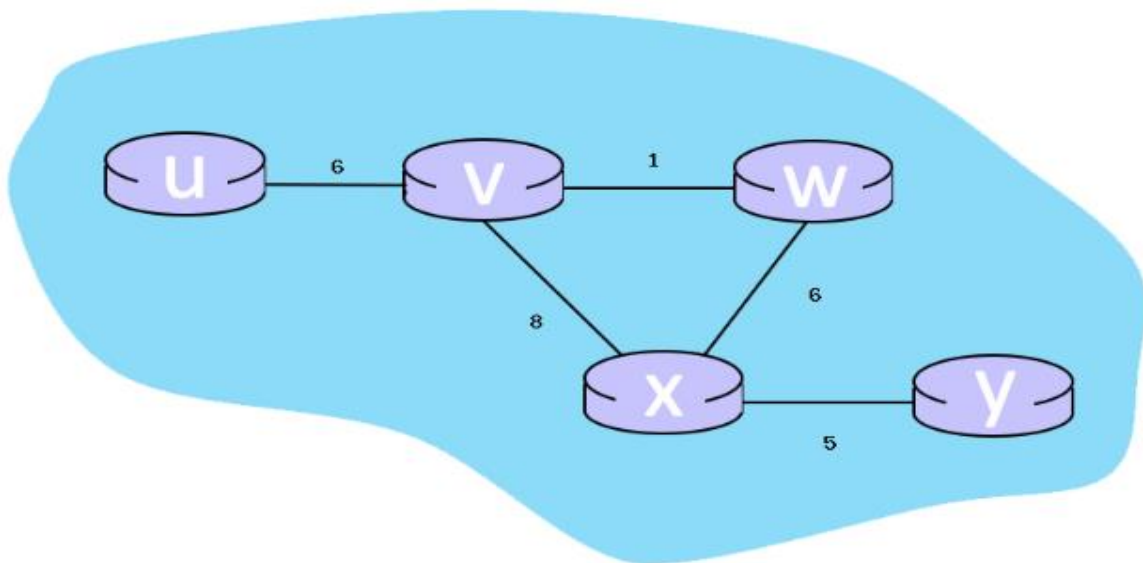


Figure 4

Part (c) A group of autonomous systems is shown in Figure 05. Answer the following questions:

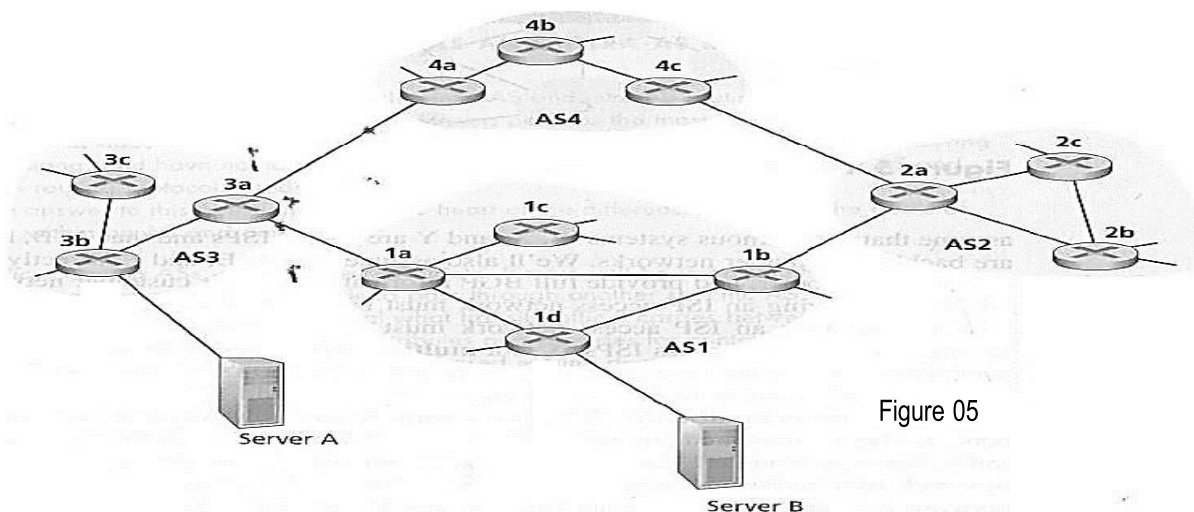


Figure 05

- Identify all BGP speaking routers in this network and label them appropriately as per your knowledge of BGP protocol.
- Using your knowledge of BGP protocol, how a PC attached to router 3b will reach Server B, if the link between router 3a and 1a is not operational.

Part (d) Consider the Figure 06 below and answer the following questions: The IP and MAC addresses are shown for nodes A, B, C and D, as well as for the router's interfaces.

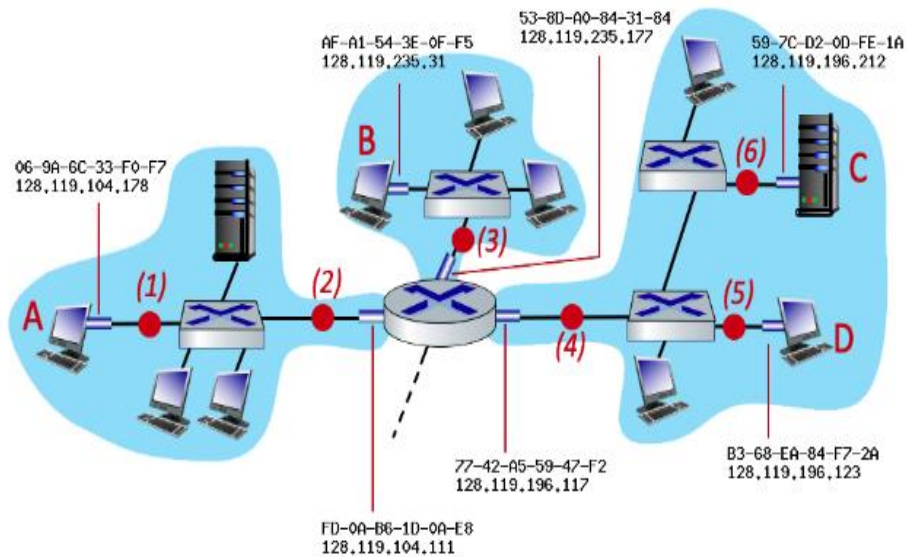


Figure 6

Consider an IP datagram being sent from node A to node D.

- What is the source mac address at point 1?
- What is the destination mac address at point 1?
- What is the source IP address at point 1?
- What is the destination IP address at point 1?

Best of Luck!!!